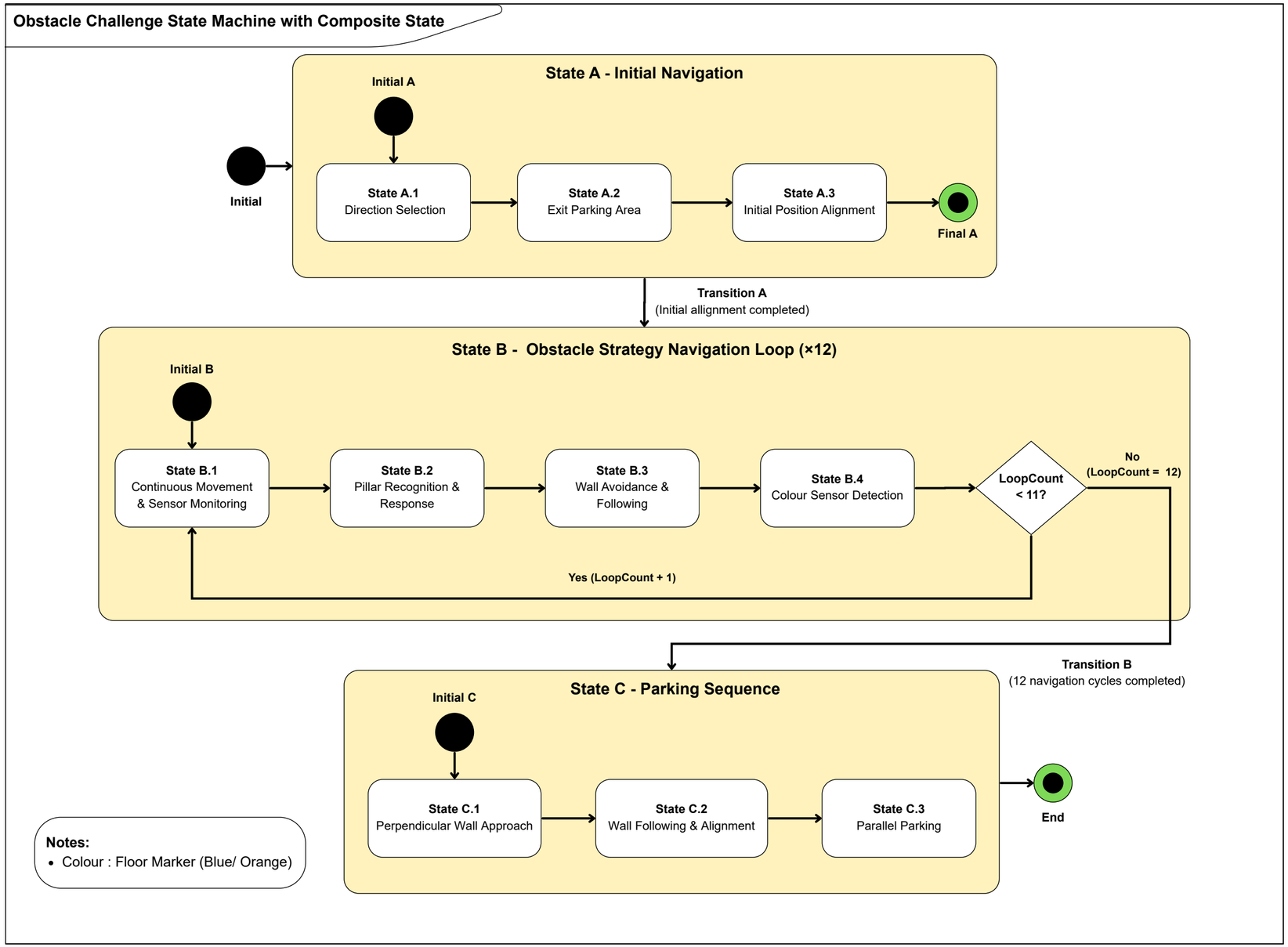

Obstacle Challenge State Machine with Composite State
State A - Initial Navigation
Initial A
Initial
State A.1
Direction Selection
State A.2
Exit Parking Area
State A.3
Initial Position Alignment
Final A
Transition A
(Initial allignment completed)
State B - Obstacle Strategy Navigation Loop (×12)
Initial B
State B.1
Continuous Movement
& Sensor Monitoring
State B.2
Pillar Recognition &
Response
State B.3
Wall Avoidance &
Following
State B.4
Colour Sensor Detection
LoopCount
< 11?
No
(LoopCount = 12)
Yes (LoopCount + 1)
Transition B
(12 navigation cycles completed)
State C - Parking Sequence
Initial C
State C.1
Perpendicular Wall Approach
State C.2
Wall Following & Alignment
State C.3
Parallel Parking
End
Notes:
Colour : Floor Marker (Blue/ Orange)